

Anqi Bao

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Summary

Applied Scientist / ML Engineer with 6+ years of experience building production ML systems in safety-critical fraud and identity domains, with hands-on experience fine-tuning LLMs, building domain-specific data pipelines, designing evaluation workflows, and deploying models for enterprise customers. Experienced in LoRA-based LLaMA-2 adaptation, transformer-based representation learning, graph neural networks, anomaly detection, and large-scale cloud ML pipelines across Azure/AWS. Seeking to advance post-training and RL challenges in reward modeling and preference optimization.

Experience

The Trade Desk

Nov 2025 - Present

Senior Applied Scientist

- Research and develop large-scale identity graph models powering downstream real-time bidding and ranking systems; analyze billions of nodes/edges to detect anomalies, evaluate graph stability, and identify emergent structure in production-scale data

Microsoft

Jun 2021 - Oct 2025

Senior Applied Scientist

- Fine-tuned LLaMA-2 with LoRA using post-training techniques for domain-specific fraud rule generation, converting natural language customer requirements into a proprietary fraud query language; owned dataset construction, prompt/label formatting, training experiments, and evaluation of generated rules against customer fraud scenarios
- Applied Direct Preference Optimization (DPO) to calibrate LLM-generated rules based on human feedback, improving rule quality and alignment with customer requirements
- Built evaluation workflows for LLM-generated fraud rules, including test-case curation, rule validity checks, precision/recall-style fraud metrics, and qualitative review with domain experts to measure whether generated outputs were deployable in real customer settings
- Engineered a pipeline to convert natural language-based features into dense embeddings using pre-trained LLMs (e.g., BERT) for fraud detection. Enhanced the pipeline with techniques such as dimensionality reduction to retain key signal components and reduce noise, resulting in measurable improvements in model accuracy and robustness
- Research and apply attention-based (Transformer) technique to extract hidden time series information from day-to-day transactional data and integrated in model to find anomaly patterns
- Designed graph neural network pipelines (DeepWalk, GAT) for large-scale entity linkage and anomaly discovery, improving detection of coordinated malicious actors. Work published in Microsoft Journal of Applied Research.
- Partnered directly with enterprise customers to understand new fraud scenarios, translate domain requirements into ML/data specifications, and build customized Azure-based data, modeling, evaluation, and visualization pipelines for model training and impact analysis

Amazon

Feb 2021 - Jun 2021

Applied Scientist

- Research, design and develop innovative production grade Machine Learning models on AWS to prevent denied parties from transacting on Amazon
- Understand the business requirements, translate them into complex analytical outputs and design tests to explain performance of models quantitatively
- Working in a highly collaborative environment partnering with various science, product management, engineering, operations, finance, business intelligence and analytics teams to develop ML systems
- Building, testing and improving prototypes given the feedback from QA and internal investigators

Microsoft

Jul 2020 - Feb 2021

Data Scientist

- Involved with entire pipeline of offline proof of concept fraud protection for potential customers. Hands on experience with Databricks (Pyspark) and Azure Pipeline for data cleaning, data mapping, feature calculation, modeling building and results visualization
- Regular Feature validation (with Cosmos DB and Scope) and model score monitor (with Power BI) for real time fraud protection for major customers

Halliburton Landmark R&D

Aug 2019 - May 2020

Software Development Engineer

- Web/cloud based UI development with Node JS and Angular JS for commercial reservoir simulator
- Couple functions in C++/JAVA/JavaScript into RESTful API backend framework
- Well control optimization with reinforcement learning methods

Halliburton Landmark

Jun 2018 - Aug 2018

Software Engineer Intern

- Computer Vision and Deep Learning Application on Wellbore Schematics using PyTorch
- Develop prototype for extracting objects information from schematics and convert it to structured data

Education

Georgia Institute of Technology

2021 - 2025

M.S., Computer Science

Texas A&M University

2015 - 2019

Ph.D., Petroleum Engineering

- GPA: 4.0 / 4.0

Texas A&M University

2017 - 2019

M.S., Statistics

- GPA: 3.8 / 4.0

Skills

- **Languages:** Python, C++, R, SQL
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Transformers, LoRA / PEFT, LLaMA-2 fine-tuning, BERT embeddings, representation learning, GNNs, anomaly detection, time-series modeling
- **Tools/Infra:** Azure, AWS, Hadoop, Spark, Kafka
- **Specialties:** Machine Learning, Deep Learning, LLM, Reinforcement Learning, Agentic AI, Post-training Techniques, Preference Optimization, Reinforcement Learning From Human Feedback (RLHF), Direct Preference Optimization (DPO)

Selected Publications

- First Author..Enhancing Fraud Detection Using Graph Embeddings.*Microsoft Journal of Applied Research, Vol 16 Article 95. Summary: DeepWalk and Graph Neural Networks (more specifically Graph Attention Network) as graph embedding techniques for generating node embeddings for real-time and batch mode evaluations. Our results demonstrate the ability for these techniques to capture essential linkage information in transactional data and provide the possibility of using the techniques in fraud protection settings.*
- First Author..Data-driven End-to-end Production Prediction of Oil Reservoirs by EnKF-enhanced Recurrent Neural Networks.*Society of Petroleum Engineering LACPEC conference. Summary: Combine Deep learning with Ensemble Kalman Filter (EnKF) to develop real-time proxy model for highly nonlinear dynamic system. An RNN model is successfully trained using time series production data. The neural network is optimized with Bayesian optimization and enhanced by EnKF for robustness.*
- Second Author..Wellbore Schematics to Structured Data Using Artificial Intelligence Tools.*Offshore Technology Conference. Summary: Use the RetinaNet model for object detection and combine with character detection tools to fully interpret and regenerate wellbore schematic. This product significantly reduces manual effort and human errors.*

Projects

Distributed Trade System

- Developed a scalable online trading platform using SpringBoot microservices, MyBatis for data access, Elasticsearch for search, RabbitMQ for messaging, Redis for caching, and deployed on AWS

Wellbore Object Detection

- Applied RetinaNet with custom character detection pipelines to automatically interpret and regenerate wellbore schematics, reducing manual effort and errors

Reservoir Simulator UI

- Developed Angular JS front end and Express API backend for Halliburton's simulator; collaborated globally using Agile practices.

Predictive Modeling

- Combined Deep learning with Ensemble Kalman Filter (EnKF) to develop real-time proxy model for nonlinear dynamic systems; optimized with Bayesian methods.